

**What can we see
when we can see
overlap?**



Modeling Homer's *Iliad*
A MNML LMNL case study with exploratory markup

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Briareus design by John Flaxman (1793), engraved by Tommaso Piroli (1795)

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Briareus
Then call'd by thee, the Monster Titan came,
Published as the Act directs. Jan 22 1795, by J. Matthews Printer Strand. (Whom Gods Briar-ars, Men Equan name.)

Pub. B. 1. 1. 522 Pope

ΡΑΨΩΔΙΑ Α

οὔνεκα τὸν Χρῦσιν ἠτίμασεν ἀρητῆρα
Ἀτρεΐδης· ὃ γὰρ ἦλθε θεὰς ἐπὶ νῆας Ἀχαιῶν
λυσόμενός τε θύγατρα φέρων τ' ἀπερείσι' ἄποινα,
στέμματ' ἔχων ἐν χερσὶν ἐκηβόλου Ἀπόλλωνος
χρυσέῳ ἀνὰ σκήπτρῳ, καὶ λίσσετο πάντας Ἀχαιοὺς,
Ἀτρεΐδα δὲ μάλιστα δύνω, κοσμήτορε λαῶν·
Ἀτρεΐδαι τε καὶ ἄλλοι εὐκνήμιδες Ἀχαιοί,
ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες
ἐκπέρσαι Πριάμοιο πόλιν, εὐ δ' οἴκαδ' ἰκέσθαι·
παῖδα δ' ἐμοὶ λύσαιτε φίλην, τὰ δ' ἄποινα δέχεσθαι,
ἄζόμενοι Διὸς υἱὸν ἐκηβόλον Ἀπόλλωνα.
ἐνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοὶ
αἰδεῖσθαι θ' ἱερῇα καὶ ἀγλαὰ δέχθαι ἄποινα·
ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,
ἀλλὰ κακῶς ἀφίει, κρατερόν δ' ἐπὶ μῦθον ἔτελλε·
μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κιχείω
ἢ νῦν δηθύνοντ' ἢ ὕστερον αὐτὶς ἰόντα,
μή νύ τοι οὐ χραίσμη σκήπτρον καὶ στέμμα θεοῖο·
τὴν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν
ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης
ἰστὸν ἐποιοχόμενῃν καὶ ἐμὸν λέχος ἀντιώωσαν·
ἀλλ' ἴθι μή μ' ἐρέθιζε σαώτερος ὥς κε νέηαι.
ὥς ἔφατ', ἔδεισεν δ' ὃ γέρον καὶ ἐπείθετο μύθῳ·
βῆ δ' ἀκέων παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·

[BOOK [n]1{}}]

[para]{prayer [to]{Calliope}}{1 [n]1.1{}}μήνιν ἄειδε θεὰ Πηληϊάδεω Ἀχιλῆος{1}
[1 [n]1.2{}}ούλομένην, ἣ μυρί' Ἀχαιοῖς ἄλγε' ἔθηκε,{1}
[1 [n]1.3{}}πολλὰς δ' ἰφθίμους ψυχὰς Ἄϊδι προΐαψεν{1}
[1 [n]1.4{}}ἡρώων, αὐτοὺς δὲ ἑλώρια τεῦχε κύνεσσιν{1}
[1 [n]1.5{}}οἰωνοῖσί τε πᾶσι, Διὸς δ' ἐτελείετο βουλή,{1}
[1 [n]1.6{}}ἔξ οὗ δὴ τὰ πρῶτα διαστήτην ἐρίσαντε{1}
[1 [n]1.7{}}Ἀτρεΐδης τε ἀναξ ἀνδρῶν καὶ δῖος Ἀχιλλεύς,{1}{prayer}{para}
[para]{1 [n]1.8{}}τίς τ' ἄρ' σφωε θεῶν ἑριδοῖ ξυνέηκε μάχεσθαι,{1}
[1 [n]1.9{}}Ἀητοὺς καὶ Διὸς υἱός· ὁ γὰρ βασιλῆϊ χολωθεὶς{1}
[1 [n]1.10{}}νοῦσον ἀνὰ στρατὸν ὄρσε κακὴν, ὀλέκοντο δὲ λαοί,{1}
[1 [n]1.11{}}οὐνεκα τὸν Χρῦσῃν ἠτίμασεν Ἀρητὴρα{1}
[1 [n]1.12{}}Ἀτρεΐδης· [scene [set]{Achaian ships}{1}ὁ γὰρ ἦλθε θοὰς ἐπὶ νῆας Ἀχαιῶν{1}
[1 [n]1.13{}}λυσόμενός τε θύγατρα φέρων τ' ἄπερείσι' ἄποινα,{1}
[1 [n]1.14{}}στέμματ' ἔχων ἐν χερσὶν ἐκηβόλου Ἀπόλλωνος{1}
[1 [n]1.15{}}χρυσῆν ἀνὰ σκήπτρῳ, καὶ λίσσετο πάντας Ἀχαιοὺς,{1}
[1 [n]1.16{}}Ἀτρεΐδα δὲ μάλιστα δῶν, κοσμήτορε λαῶν·{1}
[speech [by]{Chryses}[to]{Agamemnon, Menelaus, Achaians}{1}
[1 [n]1.17{}}Ἀτρεΐδαι τε καὶ ἄλλοι ἐϋκνήμιδες Ἀχαιοί,{1}
[1 [n]1.18{}}ὕμιν μὲν θεοὶ δοῖεν Ὀλύμπια δώματ' ἔχοντες{1}
[1 [n]1.19{}}ἐκπέρσαι Πριάμοιο πόλιν, εὖ δ' οἴκαδ' ἰκέσθαι·{1}
[1 [n]1.20{}}παῖδα δ' ἐμοὶ λύσαιτε φίλην, τὰ δ' ἄποινα δέχεσθαι,{1}
[1 [n]1.21{}}ἄζόμενοι Διὸς υἱὸν ἐκηβόλον Ἀπόλλωνα·{1}{speech}{para}
[para]{1 [n]1.22{}}ἔνθ' ἄλλοι μὲν πάντες ἐπευφήμησαν Ἀχαιοί{1}
[1 [n]1.23{}}αἰδεῖσθαι θ' ἱερῇα καὶ ἀγλαὰ δέχθαι ἄποινα·{1}
[1 [n]1.24{}}ἀλλ' οὐκ Ἀτρεΐδῃ Ἀγαμέμνονι ἦνδανε θυμῷ,{1}
[1 [n]1.25{}}ἀλλὰ κακῶς ἄφειλ, κρατερὸν δ' ἐπὶ μῦθον ἔτελλε·{1}
[speech [by]{Agamemnon}[to]{Chryses}{1}
[1 [n]1.26{}}μή σε γέρον κοίλῃσιν ἐγὼ παρὰ νηυσὶ κιχέω{1}
[1 [n]1.27{}}ἢ νῦν δηθύνοντ' ἢ ὕστερον αὖτις ἰόντα,{1}
[1 [n]1.28{}}μή νύ τοι οὐ χραίσμη σκήπτρον καὶ στέμμα θεοῖο·{1}
[1 [n]1.29{}}τὴν δ' ἐγὼ οὐ λύσω· πρὶν μιν καὶ γῆρας ἔπεισιν{1}
[1 [n]1.30{}}ἡμετέρῳ ἐνὶ οἴκῳ ἐν Ἀργεῖ τηλόθι πάτρης{1}
[1 [n]1.31{}}ἰστον ἐπιτοχομένην καὶ ἐμὸν λέχος ἀντιώσωσαν·{1}
[1 [n]1.32{}}ἀλλ' ἴθι μὴ μ' ἐρέθιζε σῴτερος ὥς κε νέηαι·{1}{speech}{para}
[para]{1 [n]1.33{}}ὥς ἔφατ', ἔδεισεν δ' ὁ γέρον καὶ ἐπειθέτο μῦθῳ·{1}{scene}
[scene [set]{seashore}{1}[1 [n]1.34{}}βῆ δ' Ἀχαιῶν παρὰ θῖνα πολυφλοίσβοιο θαλάσσης·{1}
[1 [n]1.35{}}πολλὰ δ' ἔπειτ' ἀπάνευθε κιὼν ἠρᾷθ' ὁ γεραιὸς{1}
[1 [n]1.36{}}Ἀπόλλωνι ἄνακτι, τὸν ἡὔκομος τέκε Λητώ·{1}
[prayer [to]{Apollo}[by]{Chryses}{1}
[1 [n]1.37{}}κλυθὶ μευ ἀργυρότοξ', ὃς Χρῦσῃν ἀμφιβέβηκας{1}

What are we trying to achieve? Exploratory Markup

Unrealized potentials
of declarative text encoding
Describing data with tags
regarding overlap as normal
... We can tag freely as we go ...
A schema is an object of study
(not a binding commitment)
"More like guidelines"
Markup first, hierarchies later

Depicted: TextPad® Text Editor by Helios Software Solutions

Configured for MNML LMNL

With syntax coloring, validation and operations support

Tools use **Laminator** XProc pipelines under XML Calabash or Morgana-IIIse (both Java)

The screenshot shows the TextPad editor with a file named 'book03.lmnl' open. The editor displays a mix of English and Greek text with various LMNL tags. The 'Tool Output' pane at the bottom shows the XML representation of the document, including ranges, annotations, and distinct counts for different elements like 'trope', 'tale', 'prayer', 'scene', 'set', 'simile', 'speech', 'para', and 'line'.

LMNL MYTHOI tagging

- [prayer] ... [prayer]
- [simile] ... [simile]
- [speech [by]] [to]] ... [speech]

book03.lmnl

```
167 [scene [set]Gates of Troy{}][para]
168 [1 [n]3.146{}]οἱ δ' ἀμφὶ Πρίαμον καὶ Πάνθοον ἠδὲ Θυμοίτην{1}
169 [1 [n]3.147{}]Λάμπρον τε Κλυτίον θ' Ἰκετάονά τ' ὄζον Ἀρης{1}
170 [1 [n]3.148{}]Οὐκαλέγων τε καὶ Ἀντήνων πεπνυμένω ἀμφω{1}
171 [1 [n]3.149{}]ἦατο δημογέροντες ἐπὶ Σκαίῃσι πύλῃσι,{1}
172 [1 [n]3.150{}]γῆραί δ' ἡ πολέμοιο πεπαυμένοι, ἀλλ' ἀγορηται{1}
173 [1 [n]3.151{}]ἑσθλοὶ, [simile]τεττιγεσσιν ἐοικότες οἱ τε καθ' ὕλην{1}
174 [1 [n]3.152{}]δενδρέω ἐφεζόμενοι ὅπα λειριόεσσαν ἱεῖσι.{1}
175 [1 [n]3.153{}]τοῖσι ἄρα Τρώων ἡγήτορες ἦντ' ἐπὶ πύργῳ.{1}[simile]
176 [1 [n]3.154{}]οἱ δ' ὥς οὖν εἶδονθ' Ἐλένην ἐπὶ πύργῳ ἰούσαν,{1}
177 [1 [n]3.155{}]ἦκα πρὸς ἀλλήλους ἔπεα πτερόεντ' ἀγόρευον.{1}
```

Tool Output

```
1 <report ranges="560" extent="20313">
2   <range gi="BOOK" count="1">
3     <annotation gi="n" count="1" distinct="1"/>
4   </range>
5   <range gi="trope" count="3"/>
6   <range gi="tale" count="4"/>
7   <range gi="prayer" count="5">
8     <annotation gi="to" count="5" distinct="1"/>
9     <annotation gi="by" count="4" distinct="3"/>
10  </range>
11  <range gi="scene" count="9">
12    <annotation gi="set" count="8" distinct="6"/>
13  </range>
14  <range gi="simile" count="13"/>
15  <range gi="speech" count="23">
16    <annotation gi="by" count="23" distinct="11"/>
17    <annotation gi="to" count="23" distinct="9"/>
18  </range>
19  <range gi="para" count="41"/>
20  <range gi="line" count="461">
21    <annotation gi="n" count="461" distinct="461"/>
22  </range>
23 </report>
```

Why LMNL and not another approach?

Where we have overlap, we simply mark it

All hierarchies are implicit and concurrent

Outlines emerge when nothing is forced

We could use XML out of line with pointers (standoff)

But tagging is also fun and easy

What is LMNL, anyway?

The Layered Markup and Annotation Language

Tennison and Piez, 2001 (Extreme Markup Languages)

A proposal and a research project

What if XML had no end-tag matching rule?

Laminator

<https://github.com/wendellpiez/Laminator>

How it works

XProc toolchain supports MNML

(Minimally Annotated Markup in LMNL)

LMNL, minus a couple of features

A data model reduced to essentials — or actually, two:

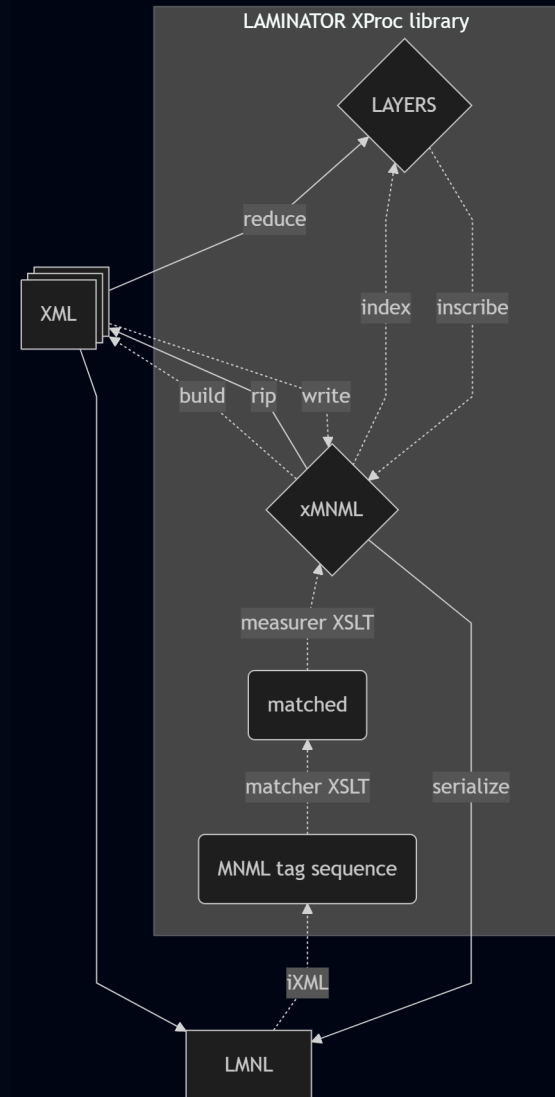
- A *tag* model (xMNML)
- And a *range* model (LAYERS)

Open-source library (XProc/XSLT) available for use and adaptation

Generic conversions are provided for

Specialized transformations and queries are easily tailored

(Easy assuming XProc with XSLT|XPath|XQuery)



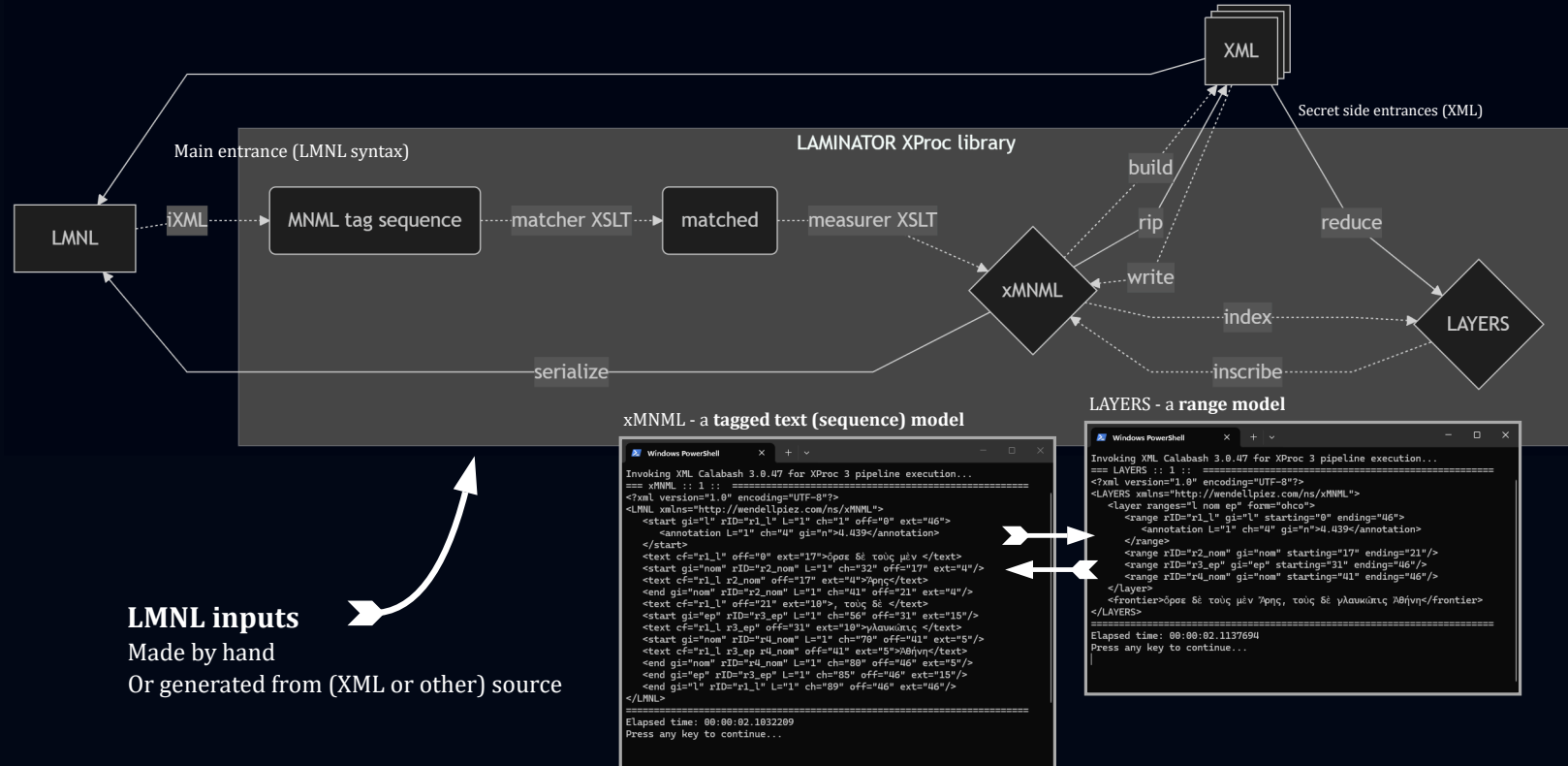
<1 n="4.439">ὄρσε δὲ τοὺς μὲν <nom>Ἀρης</nom>, τοὺς δὲ <ep>γλαυκῶπις <nom>Ἀθήνη</nom></ep></1>

<1 n="4.439">ὄρσε δὲ τοὺς μὲν <nom>Ἀρης</nom>, τοὺς δὲ γλαυκῶπις <nom>Ἀθήνη</nom></1>

XML results

One (any) hierarchy at a time

<1 n="4.439">ὄρσε δὲ τοὺς μὲν Ἀρης, τοὺς δὲ <ep>γλαυκῶπις Ἀθήνη</ep></1>



[1 [n]4.439{ }]ὄρσε δὲ τοὺς μὲν [nom]Ἀρης{nom}, τοὺς δὲ [ep]γλαυκῶπις [nom]Ἀθήνη{nom}{ep}{1}

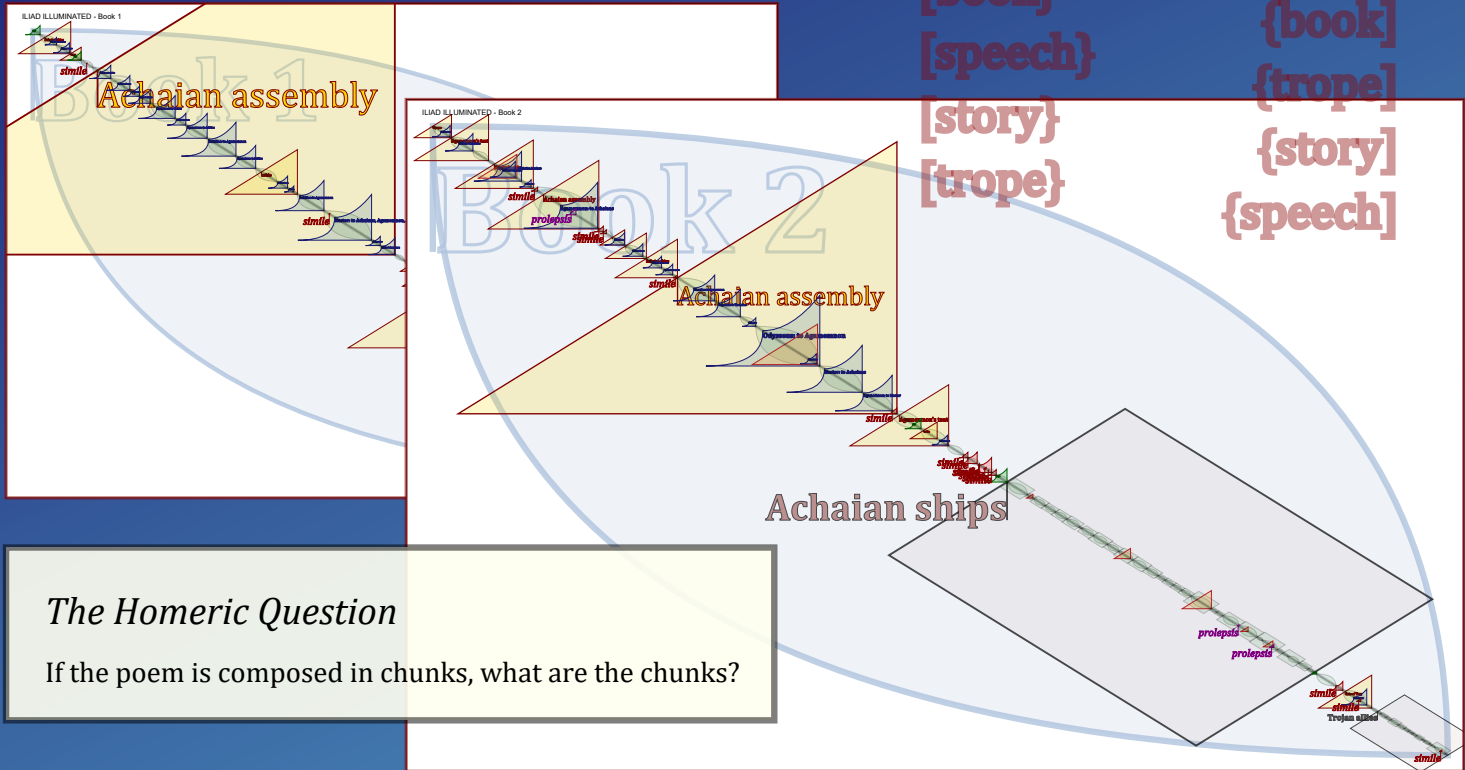
Epic Findings

The fractal structure of Homer's *Iliad*

Units of measure

morpheme, word (ονομα), line (στιχος), phalanx (φαλαγγ: ~10 lines), stratos (στρατος: ~100 lines), rhapsody (ραψωδια ~1000 lines)

Form and formula (*all the way down*) — WAVE or KYMA (κυμα) pattern



Range maps depict the extent of ranges of text within the poem.

Project goals and progress

<https://github.com/wendellpiez/EpicMarkup>

Please copy, download, fork

Technical topics

Markup, tagged text models, range models and hierarchies

Whitespace (always)

Tag (re)writing vs/and transformation

Why XProc is nice for this



Parallel initiatives (by the developer)

Using public domain data from PerseusDL and Homer Multitext Project

Homer helper site (Scriptorium Method):

<https://raventracks.org/>

Lightweight *Iliad* Assisted Reader: **Iliad Elucidated**

https://raventracks.org/#elucidated_iliad

LMNL, MNML, Laminator

LMNL markup is now easy to make

From XML or from scratch

More data, more examples?

Applications: narrative and genre studies, textual criticism and others

TODO: Formal specifications for MNML LMNL as implemented

TODO: Laminator documentation and testing

TODO: Alternate implementations including XPath 4.0 (record objects), JS, All-XSLT/iXML?

TODO: LMNL Annotations; Hyperdocument model; XML Profiles; DTS 1.0 API ...?

To come: **Iliad Illuminated**

(Some day, by small steps)

Awaiting findings first

Ideas are welcome!

For now - range map diagrams and reading notes